

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T21 · Algebraic Fractions

With Answers

Questions with full worked solutions

■ 37 pages

FAMILY MAP

Algebraic Fractions

F01	Add or subtract algebraic fractions	6 Q 17 marks
F02	Multiply or divide algebraic fractions	4 Q 16 marks
F03	Substitute or transform a rational expression	6 Q 23 marks
F04	Simplify an algebraic fraction by factor cancellation	1 Q 3 marks

Working Unit 09

4MA1-1H Q12

ORIGINAL QUESTION

3 marks

12 Express $\frac{5}{4} + \frac{x-3}{6x}$ as a single fraction in its simplest form.

(Total for Question 12 is 3 marks)

Worked Solution 09

Add algebraic fractions

UNIT 1
3 marks**Common denominator**

$$\frac{5}{4} + \frac{x-3}{6x} = \frac{15x}{12x} + \frac{2(x-3)}{12x}.$$

Combine and simplify

$$\frac{15x + 2x - 6}{12x} = \frac{17x - 6}{12x}, \quad x \neq 0.$$

Final answer

$$\frac{17x - 6}{12x}$$

Working Unit 25

4MA1-2H Q23

ORIGINAL QUESTION

4 marks

23 $\frac{4x^2 - 4x - 120}{5x^2 - 180} \div \frac{x^2 + 5x}{10x^2 + 60x} = p$ where p is an integer.

Find the value of p

Show clear algebraic working.

$$p = \dots\dots\dots$$

(Total for Question 23 is 4 marks)

Worked Solution 25

Simplify a division of algebraic fractions

UNIT 1
4 marks**Factorise**

$$\frac{4(x-6)(x+5)}{5(x-6)(x+6)} \div \frac{x(x+5)}{10x(x+6)}.$$

Multiply by the reciprocal and cancel

$$\frac{4(x-6)(x+5)}{5(x-6)(x+6)} \times \frac{10x(x+6)}{x(x+5)} = \frac{4 \cdot 10}{5} = 8.$$

Final answer

$$p = 8$$

Question 09

4MA1-1H Q12

ORIGINAL QUESTION

3 marks

(b) Express $\frac{2x+1}{4x} + \frac{7-5x}{3x}$ as a single fraction in its simplest form.

.....
(3)

Worked Solution 09

Algebraic fractions

MODULAR UNIT 1

3 marks

Use a common denominator

$$\frac{2x+1}{4x} + \frac{7-5x}{3x} = \frac{3(2x+1) + 4(7-5x)}{12x} = \frac{31-14x}{12x}.$$

Final answer

$$\frac{31-14x}{12x}$$

Question 10

4MA1-1H Q11 demand

ORIGINAL QUESTION UNIT 1

3 marks

- 11 Express $\frac{3}{4} + \frac{5-x}{6x}$ as a single fraction in its simplest terms.

.....

Worked solution 10

Single algebraic fraction

MODULAR UNIT 1

3 marks

Demand – Single algebraic fraction

Use a common denominator

$$\frac{3}{4} + \frac{5-x}{6x} = \frac{9x + 2(5-x)}{12x}.$$

Simplify

$$\frac{9x + 10 - 2x}{12x} = \frac{7x + 10}{12x}.$$

Final answer $(7x + 10)/(12x)$

Question 31

4MA1-2H Q24 whole

ORIGINAL QUESTION UNIT 1

5 marks

24 Solve $\frac{45x^2 - 80x}{3x^2 + x - 4} \times \left(\frac{1}{3x - 4} + \frac{1}{x} \right) = \frac{4(x + 2)}{5x - 8}$

Show clear algebraic working.

$x =$

Worked solution 31

Solve a rational algebraic equation

MODULAR UNIT 1

5 marks

whole – Solve a rational algebraic equation

Combine the bracketed fractions

$$\frac{1}{3x-4} + \frac{1}{x} = \frac{x+3x-4}{x(3x-4)} = \frac{4(x-1)}{x(3x-4)}.$$

Factor the first fraction

$$45x^3 - 80x = 5x(3x-4)(3x+4), \quad 3x^2 + x - 4 = (3x+4)(x-1).$$

Cancel common factors

The left-hand side simplifies to 20 (with the original denominator restrictions retained).

Solve the remaining equation

$$20 = \frac{4(x+2)}{5x-8} \Rightarrow 20(5x-8) = 4(x+2).$$

Find and check x

$$96x = 168 \Rightarrow x = \frac{7}{4}, \text{ which is not an excluded denominator value.}$$

Final answer

$$x = 7/4$$

Original question 27

4MA1-2H Q23 M01

MODULAR UNIT 1

4 marks

23 Show that $\frac{16x^2 - 36}{x - 7} \div \frac{2x^2 + 7x + 6}{x^2 - 5x - 14} - (7 + 8x) = n$

where n is an integer to be found.

Show clear algebraic working.

Worked solution 27

Simplify an algebraic fraction

MODULAR UNIT 1

4 marks

M01 – Simplify an algebraic fraction**Factorise the first numerator**

$$16x^2 - 36 = 4(2x - 3)(2x + 3).$$

Factorise the remaining terms

$$2x^2 + 7x + 6 = (2x + 3)(x + 2), \quad x^2 - 5x - 14 = (x - 7)(x + 2).$$

CancelThe quotient simplifies to $4(2x-3)$.**Subtract the linear expression**

$$4(2x-3)-(7+8x)=-19, \text{ so } n=-19.$$

Final answer

$$n = -19$$

Original question 08

4MA1-1H Q11 Q11(b)

ORIGINAL QUESTION**3 marks**

(b) Express $\frac{7}{8} - \frac{x+3}{4x}$ as a single fraction in its simplest form.

.....
(3)

Worked solution 08

Chapter: Algebraic Fractions

MODULAR UNIT 1

3 marks

Q11(b) – Chapter: Algebraic Fractions

Family: Add or subtract algebraic fractions • Variant: Linear denominators

MethodUse denominator $8x$: $\frac{7}{8} - \frac{(x + 3)}{(4x)} = \frac{7x}{(8x)} - \frac{2(x + 3)}{(8x)}$.*Why: $8x$ is a common denominator for 8 and $4x$.***Method**Combining gives $\frac{[7x - 2(x + 3)]}{(8x)} = \frac{(7x - 2x - 6)}{(8x)}$.*Why: Keep the subtraction outside the bracket when combining numerators.***Method**The single simplified fraction is $\frac{(5x - 6)}{(8x)}$.*Why: Collect the numerator terms.***Final answer****single fraction:** $\frac{(5x - 6)}{(8x)}$

Original question 26

4MA1-2H Q19 Q19

ORIGINAL QUESTION**3 marks**

19 $a = \frac{14}{3x - 7} \quad x = \frac{7}{4y - 3}$

Express a in the form $\frac{py + q}{ry + s}$ where p, q, r and s are integers.

Give your answer in its simplest form.

$a = \dots\dots\dots$

Worked solution 26

Chapter: Algebraic Fractions

MODULAR UNIT 1

3 marks

Q19 – Chapter: Algebraic Fractions

Family: Substitute or transform a rational expression • Variant: Substitute a variable relation into a fraction

MethodSubstitute $x = 7/(4y - 3)$: $a = 14/[21/(4y - 3) - 7]$.*Why: Replace x in the definition of a .***Method** $a = 14(4y - 3)/[21 - 7(4y - 3)] = 14(4y - 3)/(42 - 28y)$.*Why: Combine the denominator into one fraction.***Method**Cancel the common factor 14: $a = (4y - 3)/(3 - 2y)$.*Why: Factor and simplify completely.***Final answer****a:** $a = (4y - 3)/(3 - 2y)$

24 Express

$$\left(\frac{4}{2x-5} - \frac{3}{2x-3} \right) \div \frac{9x-4x^3}{6x^2-17x+5}$$

as a single fraction in its simplest form.

Worked solution

January 2020 | Question 24 | 4 marks

Family: Multiply or divide algebraic fractions • Variant: Divide factorisable algebraic fractions

Method / working

1. Simplify the first bracket to $(2x+3)/[(2x-5)(2x-3)]$.
2. Factor the divisor numerator and denominator.
3. Multiply by the reciprocal and cancel common factors.
4. Final fraction= $(1-3x)/[x(2x-3)^2]$.

Final answer: $(1-3x)/(x(2x-3)^2)$

- 12 (a) Express $\frac{4}{x-2} - \frac{3}{x+1}$ as a single fraction.
Give your answer in its simplest form.

(3)

Worked solution

January 2021 | Question 12 | 3 marks

Family: Add or subtract algebraic fractions • Variant: Linear denominators

Method / working

1. Use common denominator $(x-2)(x+1)$.

2. Numerator = $4(x+1)-3(x-2)=x+10$.

3. State the simplified fraction.

Final answer: $(x+10)/((x-2)(x+1))$

14 (a) Solve $\frac{9a - 7}{5} - \frac{3a - 7}{4} = 4.55$

Show clear algebraic working.

$a = \dots\dots\dots$
(3)

Worked solution

January 2021 | Question 14 | 3 marks

Family: Substitute or transform a rational expression • Variant: Equation containing algebraic fractions

Method / working

1. Multiply the equation by 20 to clear denominators.
2. Expand to obtain $21a=84$.
3. Therefore $a=4$.

Final answer: $a=4$

21 Write $\frac{25x^2 - 64}{5x^2 - 13x - 6} \times \frac{x^2 - 8x + 15}{5x + 8} - (x - 7)$

as a single fraction in its simplest form.
Show clear algebraic working.

Worked solution

January 2021 | Question 21 | 4 marks

Family: Multiply or divide algebraic fractions | Variant: Multiply factorisable algebraic fractions

METHOD / WORKING

1. Factor $25x^2 - 64 = (5x - 8)(5x + 8)$, $x^2 - 8x + 15 = (x - 5)(x - 3)$, and $5x^2 - 13x - 6 = (5x + 2)(x - 3)$.
2. Factor and cancel common factors before combining.
3. Put the remaining terms over a common denominator.
4. The simplified single fraction is $54/(5x+2)$.

FINAL ANSWER

 $54/(5x+2)$

21 Given that $x = \frac{5}{9y+5}$ and that $y = \frac{5}{5a-2}$

find an expression for x in terms of a .

Give your expression as a single fraction in its simplest form.

Worked solution

June 2021 | Question 21 | 4 marks

Family: Substitute or transform a rational expression • Variant: Substitute a variable relation into a fraction

Method / working

1. Use the printed relations $x=5/(9y+5)$ and $y=5/(5a-2)$.
2. Substitute y : $9y+5 = 45/(5a-2)+5 = [45+5(5a-2)]/(5a-2)$.
3. Simplify the denominator to $5(5a+7)/(5a-2)$, then divide 5 by this fraction.
4. Therefore $x=(5a-2)/(5a+7)$.

Final answer: $(5a-2)/(5a+7)$

(b) Write as a single fraction $\frac{2}{3x} + \frac{4}{5x} - \frac{9}{10x}$

Give your answer in its simplest form.

(2)

Worked solution

November 2021 | Question 12 | 2 marks

Family: Add or subtract algebraic fractions | Variant: Linear denominators

METHOD / WORKING

1. (b)

$$2. \frac{2}{3x} + \frac{4}{5x} - \frac{9}{10x}$$

$$3. = \frac{20 + 24 - 27}{30x} = \frac{17}{30x}$$

FINAL ANSWER

$$(b) \frac{17}{30x}$$

21 Express

$$\frac{1}{3x-2} \times \frac{9x^2-4}{3x^2-13x-10} - \frac{7}{x-1}$$

as a single fraction in its simplest form.

(Total for Question 21 is 5 marks)

Answer reference | May-Nov 2020 | Question 21

Family: Substitute or transform a rational expression • Variant: Substitute a variable relation into a fraction

21	$\left(\frac{9x^2 - 4}{3x^2 - 13x - 10} = \right) \frac{(3x+2)(3x-2)}{(3x+2)(x-5)}$			M1 for either $(3x+2)(3x-2)$ or $(3x+2)(x-5)$	M2 for $\frac{9x^2 - 4}{(9x^2 - 4)(x-5)} =$ $\frac{1}{(x-5)}$
	$\left(\frac{9x^2 - 4}{3x^2 - 13x - 10} = \right) \frac{(3x+2)(3x-2)}{(3x+2)(x-5)}$			M1 for $(3x+2)(3x-2)$ and $(3x+2)(x-5)$	
	E.g. of denominators $(3x-2)(3x^2-13x-10)(x-1)$ or $(3x-2)(3x+2)(x-5)(x-1)$ or $9x^4 - 54x^3 + 41x^2 + 24x - 20$ or $(3x+2)(x-5)(x-1)$ or $3x^3 - 16x^2 + 3x + 10$ or $(3x-2)(x-5)(x-1)$ or $3x^3 - 20x^2 + 27x - 10$ or $(x-5)(x-1)$ or $x^2 - 6x + 5$			M1 (indep) ft their fractions for use of a correct common denominator for 2 fractions with algebraic denominators NB: fractions need not be simplified	
	$\frac{x-1-7(x-5)}{(x-5)(x-1)}$ or $\frac{x-1-7x+35}{(x-5)(x-1)}$ or $\frac{x-1-7(x-5)}{x^2-6x+5}$ or $\frac{x-1-7x+35}{x^2-6x+5}$ oe			M1 for a correct fraction with a correct quadratic denominator – may or may not be expanded which leads to a correct answer	
		$\frac{2(17-3x)}{(x-5)(x-1)}$	5	A1 accept $\frac{34-6x}{(x-5)(x-1)}$ oe; if denominator is expanded then it must be correct	
Total 5 marks					

Worked solution

May-Nov 2020 | Question 21 | 5 marks

Family: Substitute or transform a rational expression | Variant: Substitute a variable relation into a fraction

METHOD / WORKING

Worked solution
May-Nov 2020 Question 21 5 marks
Family: Substitute or transform a rational expression Variant: Substitute a variable relation into a fraction
Method
$\frac{13x - 2x(9x^2 - 4)}{(3x^2 - 13x - 10)}$
$-\frac{7}{x-1}$
Factor:
$9x^2 - 4 = (3x - 2)(3x + 2)$
$3x^2 - 13x - 10 = (3x + 2)(x - 5)$
So the first product becomes
$\frac{1}{(3x-2)} \times$
$\frac{((3x-2)(3x+2))}{((3x+2)(x-5))}$
Cancel common factors:
$\frac{1}{x-5}$
Now subtract:
$\frac{1}{x-5} - \frac{7}{x-1}$
Use the common denominator $(x-5)(x-1)$:
$\frac{(x-1-7(x-5))}{((x-5)(x-1))}$
$= \frac{(x-1-7x+35)}{((x-5)(x-1))}$
$= \frac{(34-6x)}{((x-5)(x-1))}$
Answer: $\boxed{\frac{(34-6x)}{((x-5)(x-1))}}$

FINAL ANSWER

13 Solve $\frac{x+3}{4} - \frac{7-x}{5} = 4.3$

Show clear algebraic working.

$x =$

(Total for Question 13 is 3 marks)

Answer reference | January 2023 | Question 13

Family: Simplify an algebraic fraction by factor cancellation • Variant: Common monomial cancellation

Question	Working	Answer	Mark	Notes
13	eg $20 \times \frac{x+3}{4} - 20 \times \frac{7-x}{5} = 20 \times 4.3$ or eg $5(x+3) - 4(7-x) = 20 \times 4.3$ or eg $\frac{5(x+3)}{20} - \frac{4(7-x)}{20} (= 4.3)$ or eg $\frac{5(x+3) - 4(7-x)}{20} (= 4.3)$		3	M1 For clear intention to multiply all terms by 20 (or 4×5) or a multiple of 20 oe or to express LHS as two fractions over 20 (or 4×5) or a multiple of 20 oe or as a single fraction with a denominator of 20 (or 4×5) or a multiple of 20 oe if expanded numerator, allow one error
	eg $5x + 15 - 28 + 4x = 4.3 \times 20$ oe eg $9x - 13 = 86$ eg $9x = 99$			M1 Expanding brackets and multiplying by denominator with no more than one error in total from multiplying out brackets [we must see 4.3×20 or 86 accurately]
	<i>Working required</i>	11		A1 dep on M1
Total 3 marks				

Worked solution

January 2023 | Question 13 | 3 marks

Family: Simplify an algebraic fraction by factor cancellation • Variant: Common monomial cancellation

Official final mark-scheme pages are embedded immediately after this question.

23 $a = \frac{2x+5}{1-x} \quad x = \frac{5-2y}{3y}$

Write a in the form $\frac{m+ny}{p(y-1)}$ where m , n and p are integers.

Show your working clearly.

$a = \dots\dots\dots$

(Total for Question 23 is 3 marks)

Worked solution

November 2025 | Question 23 | 3 marks

Family: Substitute or transform a rational expression | Variant: Substitute a variable relation into a fraction

METHOD / WORKING

Worked solution
November 2025 Question 23 3 marks
Family: Substitute or transform a rational expression Variant: Substitute a variable relation into a fraction
Method
$a = (2x+5)/(1-x), \text{quad } x = (5-2y)/(3y)$
Substitute for x.
Numerator:
$2x+5 = 2((5-2y)/(3y))+5$
$= (10-4y+15y)/(3y)$
$= (10+11y)/(3y)$
Denominator:
$1-x = 1-(5-2y)/(3y)$
$= (3y-5+2y)/(3y)$
$= (5y-5)/(3y)$
$= (5(y-1))/(3y)$
Therefore
$a = ((10+11y)/(3y))/((5(y-1))/(3y))$
$= (10+11y)/(5(y-1))$
Answer: $d^{\frac{10+11y}{5(y-1)}}$
PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION

FINAL ANSWER